



BBIT/INTE 423

BUSINESS PROCESS REENGINEERING

Lecture 1: Introduction: Why BPR

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Outline

- **Demystifying the Key terms**
- **Digital Darwinism**
- **Linking the key terms for business application**
- **The differences**
- **Case study**



Digital Darwinism: How Disruptive Technology Is Changing Business for Good

- Digital Darwinism — a phenomenon where technology and society are evolving faster than businesses can naturally adapt. This sets the stage for a new era of leadership, a new generation of business models, charging behind a mantra of “adapt or die.”
- Rather than react to change or be disrupted by it, some forward-looking companies are investing in **digital transformation** to adapt and outperform peers.
- **digital transformation** is the process of leveraging **digitization** to transform business outcomes
- Business process reengineering is a step towards **digital transformation**



Automation, digitization, computerization and Digitalization

Word	Definition
Automation	Automatically controlled operation of an apparatus, process, or system by mechanical or electronic devices that take the place of human labor.
Digitization	the conversion of text, pictures, or sound into a digital form that can be processed by a computer.
Computerization	the process of starting to use a computer to do something that was done by people or other machines before.
Digitalization	the use of digital technologies to change a business model and provide new revenue and value-producing opportunities; it is the process of moving to a digital business.
Bonus: Digital Technology	electronic tools, systems, devices and resources that generate, store or process data.

Source: Mirriam-Webster, Oxford Language, Cambridge Dictionary, Gartner, Victoria State Government



Linking Them all Together for Business Productivity

➤ Technically:

- ❖ you could digitize without computerizing but there is no need
- ❖ You need to digitize to computerize, since digitization is just getting data and information into a format that computers can process.
- Computerization is modern automation using computers. It replaces non-computerized automation.
- Digitalization takes all of these building blocks and combines them to harness the full potential of computerization and digital technologies.
- Digitalization further leverages digital technology to help you make data-backed business decisions. This technology allows you to gather, link, and analyze large amounts of precisely-collected data which enables these informed decisions



Examples of digitization:

Writing notes on paper and then typing them out into a Word file or an Excel spreadsheet.

Scanning a document and saving it on the computer's hard drive as a digital file in a format such as PDF.

Converting analog VHS cassettes to DVD, CD, or Blu-ray discs that include additional data

Examples of digitalization:

Uploading a PDF file from the computer's hard drive to the cloud so that it can be shared with many people at once and analyzed using yet another tool based in the cloud.

Uploading digital movie files from CD, DVD, or Blu-ray discs to online services to allow people to download, stream, or rent them online.

Converting a Word document or Excel spreadsheet to a file in Google Docs or Google Sheets stored in Google's cloud platform - that's how the document can be shared among several users at once.



Digitization, Digitalization and Digital Transformation

- These three terms are constantly used to talk about how businesses embrace modern technologies today.
- The three concepts end up sounding very similar - no wonder that they're sometimes used interchangeably. But in reality, they refer to completely different activities with different purposes, goals, and implications on business operations.
- Digitalization is a process that takes advantage of digitization in order to improve business processes i.e.
- Digitalization means putting the digitized information to work.



Digitization, Digitalization and Digital Transformation

- Digital transformation is the last stage in this journey
- It is a process that takes advantage of both digitization and digitalization.
- digital transformation uses digital technologies in order to create new or modify the existing business models, processes, cultures, and customer experiences.
- The goal is to Address the changing business and market requirements as new technologies permeates the business environment
- ***Digital transformation is all about reinventing the business for the digital age. It's a customer-centric process - it starts with how a company thinks about and engages with customers.***



Digitization, Digitalization and Digital Transformation

- **digital transformation** is the process of leveraging **digitization** to transform business outcomes
- Its about getting more innovative to maximize business benefits from the new technologies
- From Digitization (get digital content) → use the digital content to improve business process (digitalization) → transform business outcomes by getting more innovative (digital transformation)



Digital transformation case study – Netflix

- <https://producthabits.com/how-netflix-became-a-100-billion-company-in-20-years/> ---- complete story here
- Netflix started out as a mail-order service for renting videos?
 - ❖ This move already disrupted the sector, which back then was based on brick-and-mortar video rental stores.
 - ❖ ***Service Delivery innovation 1:*** started with a frictionless DVD rental business facilitated by the internet
 - ❖ ***Service delivery Innovation 2:*** As digital technologies evolved, streaming video online became possible – and [Netflix instantly jumped on that opportunity](#), developing a streaming platform used all over the world. Users can stream video content directly but also get personalized recommendations that are perfectly in line with their thanks to the AI recommendation engine.
 - ❖ developed an entirely new streaming business from scratch where customers could watch as much as they want, as opposed to paying per rental



Digital transformation case study – Netflix

- Today, Netflix assumes the role of a traditional broadcast and cable TV network and production studio all at once as it develops a library of on-demand content at competitive prices.
- **Other case studies:**
 - ❖ Fujifilm leveraged on the digital technology to leapfrog in the film industry
 - ❖ Contrary to Fujifilm Kodak became bankrupt for failure to embrace digital technologies and transform as market evolved



Types of Digital Transformation

➤ **Process Transformation.**

- ❖ Involves radical changes to business processes to adapt to new conditions and requirements and meet business goals.
- ❖ business process transformation is a way of creating a strategic advantage that sets up a company for long-term success and short-term dividends.

➤ **Business Model Transformation.**

- ❖ Business model transformation aims to fundamentally change the way companies deliver value for customers. For example, Netflix made the switch from mailing DVDs to online streaming, while Blockbuster failed to make the transition, which ultimately led to the latter company's eventual fate.
- ❖ reshaping go to market strategy to suite evolving customer needs. Customers often signal their readiness for a different type of relationship through their purchasing patterns.



Types of Digital Transformation

➤ Domain Transformation. .

- ❖ Organizations typically adopt new technologies to redefine their products and services. They might extend current services to a new customer base or develop entirely new technology-enabled offerings. For example, offering equipment on a rental basis rather than purchase-only enables you to reach a previously unserved segment of customers who may only need your technology temporarily.
- ❖ Taking advantage of emerging digital technologies to Venture into new business frontiers
- ❖ ***E.g. a university can transform is knowledge delivery business model by recording quality lessons for sale through Udacity***



Types of Digital Transformation

➤ Cultural/Organizational Transformation.

- ❖ Work culture change to Embrace a digital-first culture enables organizations to adopt agile workflows, develop a bias toward testing and learning as well as support decentralized decision-making.
- ❖ A typical mindset shift alters from managing production output to focusing on customer service and innovation. Often, the cultural shift occurs organically during other transformation initiatives as internal teams adopt digital workflows and recognize the power of changing organizational norms.



Business Process Reengineering vs. Digital Transformation

- BPR is a continuous process that empowers an organization to move beyond traditional operations and embrace Digital Transformation.
- In an enterprise's roadmap towards Digital Transformation and operational efficiency, BPR is the keystone. Here are four ways through which BPR aids the journey towards digital revolution:
 1. **Streamlining Processes**
 2. **Business Process Automation:** incorporating digital tools to transform their existing business processes through Business Process Automation (BPA). This involves deploying a fully automated, flexible process to manage existing and new processes.
 3. **Leveraging Analytics:** Analytics can be used for elementary reporting and business intelligence (Descriptive Analytics) to prescribe the best possible solution or alternatives (Prescriptive Analytics), as well as predict outcomes (Predictive Analytics). It is worthwhile learning from digitally-enabled businesses that use advanced analytics like Artificial Intelligence, Deep Learning, and NLP for various insights.



Business Process Reengineering vs. Digital Transformation

4. Big Data Processing

- ❖ Presently, almost all businesses hold a huge amount of data, both structured and unstructured. This data in conjunction with analytical tools can lead to the harvesting of valuable insights and supporting business decisions. Big data is all about combining structured and unstructured data to get fresh insights that businesses didn't have access to in the previous years